Monitoring biodiversity for human, animal, plant and environmental health

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Abstract: The One Health approach promotes collaboration across disciplines to enhance the health of humans, animals, plants, and the environment. The Quadripartite organizations, which include the Food and Agriculture Organization of the United Nations (FAO), the United Nations Environment Programme (UNEP), the World Organisation for Animal Health (WOAH), and the World Health Organization (WHO), developed the One Health Joint Plan of Action (OH JPA) to support countries in achieving One Health. This plan consists of six action tracks, each consisting of a set of actions for implementing One Health. By requiring knowledge on zoonotic diseases (tracks 2 and 3), food and agriculture (track 4), antimicrobial resistance (track 5), and environmental health (track 6), most of these tracks directly concern biodiversity. However, there are currently no indicators for monitoring the OH JPA. Our research examines the extent to which all six tracks are covered by the Kunming-Montreal Global Biodiversity Framework (KM-GBF) of the Convention on Biological Diversity (CBD), which contains many indicators at the intersection of biodiversity and health. We assessed (1) the link between each indicator of the KM-GBF and human, animal, plant, and environmental health and (2) the usability of these indicators for monitoring One Health actions. We found that 75% of indicators are associated with health, and that a similar proportion can be used for monitoring One Health actions. Overall, our work aims to strengthen collaboration between the CBD Secretariat and the Quadripartite Organizations by highlighting the need for shared data, policy, and governance practices.

Keywords: biodiversity indicators, Kunming-Montreal Global Biodiversity Framework, One Health, One Health Joint Plan of Action, Quadripartite Organizations

Introduction

- The One Health approach
  - Interconnection between human, animal, plant, and environmental health
  - Zoonotic diseases, non-communicable diseases, food safety, antimicrobial and antiparasitic resistance, climate change, pollution
  - Collaboration across disciplines
- The One Health Joint Plan of Action
  - Quadripartite Organizations
  - 6 action tracks, many actions, even more activities
  - No indicators
- The Kunming-Montreal Global Biodiversity Framework
  - Convention on Biological Diversity
  - Protecting biodiversity by working towards targets and goals
  - Global Action Plan recognizes that biodiversity is linked with health
- Monitoring framework of the KM-GBF
  - Types of indicators (headline, binary, component, complementary)
  - Many indicators are linked with health (examples)
  - Reusing indicators decreases the workload of countries
- Objectives of our study
  1. Assess the link between biodiversity indicators and human, animal, plant, and environmental health
     - Strengthens the link between biodiversity and health
     - Reinforces the need for collaboration across disciplines
     - Highlights the need for shared policy and governance practices between the CBD Secretariat and the Quadripartite Organizations
  2. Evaluate the usability of indicators for monitoring One Health actions
     - Highlights the need for data sharing between Parties, organizations, and other stakeholders
     - Reduces the workload on countries

Methods

We qualitatively assessed the relevance of all 204 KM-GBF indicators (Conference of the Parties to the Convention on Biological Diversity, 2025) to Health. Specifically, we evaluated the degree to which they were linked to each of the four pillars of One Health (i.e., to human, animal, plant, and environmental heath), as well as their usability for monitoring the OH JPA. Our assessments were informed by the metadata (rationale, definition, method of computation, data sources, and scale of use) of the indicators provided by the World Conservation Monitoring Centre of the UN Environment Programme (UNEP-WCMC, 2025), when available, looking for explicit considerations of Health. When such metadata was unavailable, assessments were entirely based on expert judgment. The result of this analysis and the code to reproduce the figures are available on Zenodo.

Assessing the link between indicators and health

- Qualitative assessments
  - Assessing the link between each indicator and human, animal, plant, and environmental health
  - Direct connection if there is a direct causal relationship between the indicator and health (e.g., the indicator could directly measure the state or a risk factor of health)
  - Indirect connection if there is a single intermediary factor between the indicator and health
  - Potential connection if there are two or more intermediary factors between the indicator and health, or if they are likely connected but we are not sure how
  - No connection if the connection between the indicator and health is far-fetched, unlikely, or absent
  - Require categorizing species and defining health
- Categorizing species within One Health
  - Animals
    - Include pets, livestock, fisheries, and aquaculture, i.e. species currently looked after by veterinarians and food inspectors
    - Exclude humans and wildlife
    - Are taken care of by the World Organisation for Animal Health (WOAH)
  - Humans
    - Are taken care of by the World Health Organization (WHO)

    - Plants
        - Include species used for food, fuel, and medicine, i.e. cultivates plants
        - Are taken care of by the Food and Agriculture Organization of the United Nations (FAO)
    - Environment
        - Includes ecosystems and all species not considered in the human, animal, or plant categories
        - Includes forestry and fisheries
        - Being taken care of by the United Nations Environment Programme (UNEP)
- Defining health
    - Human and animal health
        - Overall wellbeing of an individual, i.e. the extent to which it is able to function physically, mentally, and behaviorally
        - Diseases are deviations from the normal functioning of an individual, often leading to pain, suffering, and death
    - Plant health
        - The extent to which an individual is able to function physically
        - Diseases are deviations from the normal physiological functioning of an individual, often leading to death
    - Environmental health
        - The extent to which the environment is able to function, maintain biological and chemical processes, and adapt to change
        - Disturbances are degradations that lead to a decline in the functioning of ecosystems and biological communities
        - Environmental health include wildlife health

Assessing the usability of indicators for monitoring the OH JPA

- Qualitative assessments
    - Evaluating each action track independently
    - Identifying the main action that can be monitored for each relevant action track
    - Directly usable indicators can already be used to monitor an action in the action track

- Indicators usable after adaptation need to be slightly modified (e.g., changes in scale of measurement, data resolution, or taxa) before being used to monitor an action in the action track
- Not usable indicators need to be greatly modified before being used to monitor the actions in the action track, or they monitor something outside the scope of the action track

Results and discussion

Link between biodiversity indicators and health

- Most indicators are linked with health
  - How many indicators are directly or indirectly linked with human, animal, plant, and environmental health?
  - Description of Figure 1
  - Examples of indicator linked with human, animal, plant, and environmental health
- Link between biodiversity and health
  - The state of biodiversity impacts health
  - Ecosystem services benefit health
  - Biodiversity and health have similar pressures
  - Biodiversity and health are protected with similar actions

Usability of indicators for monitoring the OH JPA

- Most indicators can be used to monitor the OH JPA
  - How many indicators for each action track?
  - Description of Table 1
  - Description of Figure 2
  - Importance of directly reusing indicators
  - Indicators usable after adaptation are based on similar and robust methodologies, which minimizing training requirements
- Many usable indicators are headline and binary indicators
  - How many?
  - Presentation of important gaps
  - Description of Figure 3

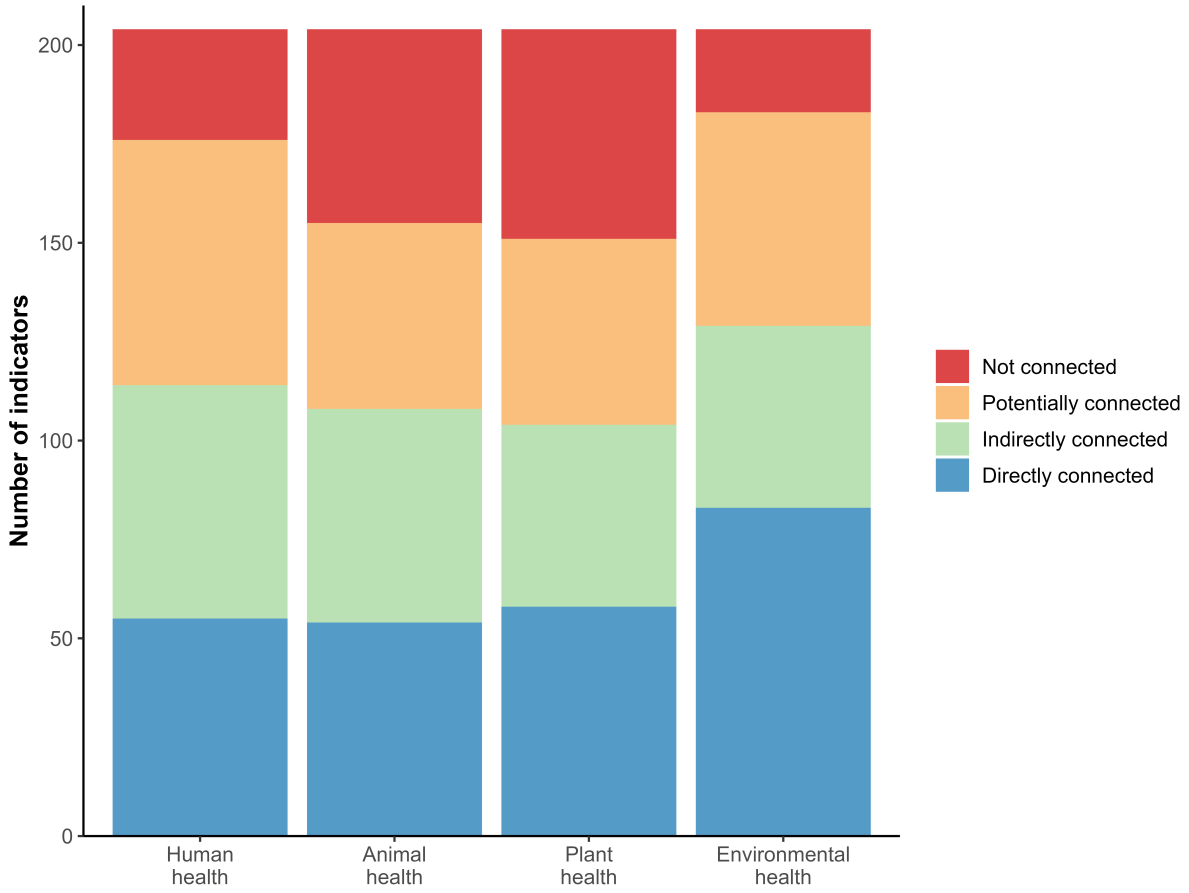

Figure 1: **Number of KM-GBF indicators linked with human, animal, plant, and environmental health.** Each indicator was classified as being either directly connected (blue), indirectly connected (green), potentially connected (orange), or not connected (red) to each of the Health dimensions.

    - Important because these are mandatory indicators that are more likely to be measured
- Usable indicators are in all categories of the Action Plan
    - The KM-GBF addresses many dimensions of health
    - Presentation of important gaps
    - Description of important categories and their connection with health

Conclusion

- Monitoring Frameworks
    - The monitoring framework of the OH JPA can be based on indicators of the KM-GBF
    - Importance of reusing indicators to decrease workload on countries
    - Importance of sharing data, methodologies, and expertise
    - Sharing policy and governance practices

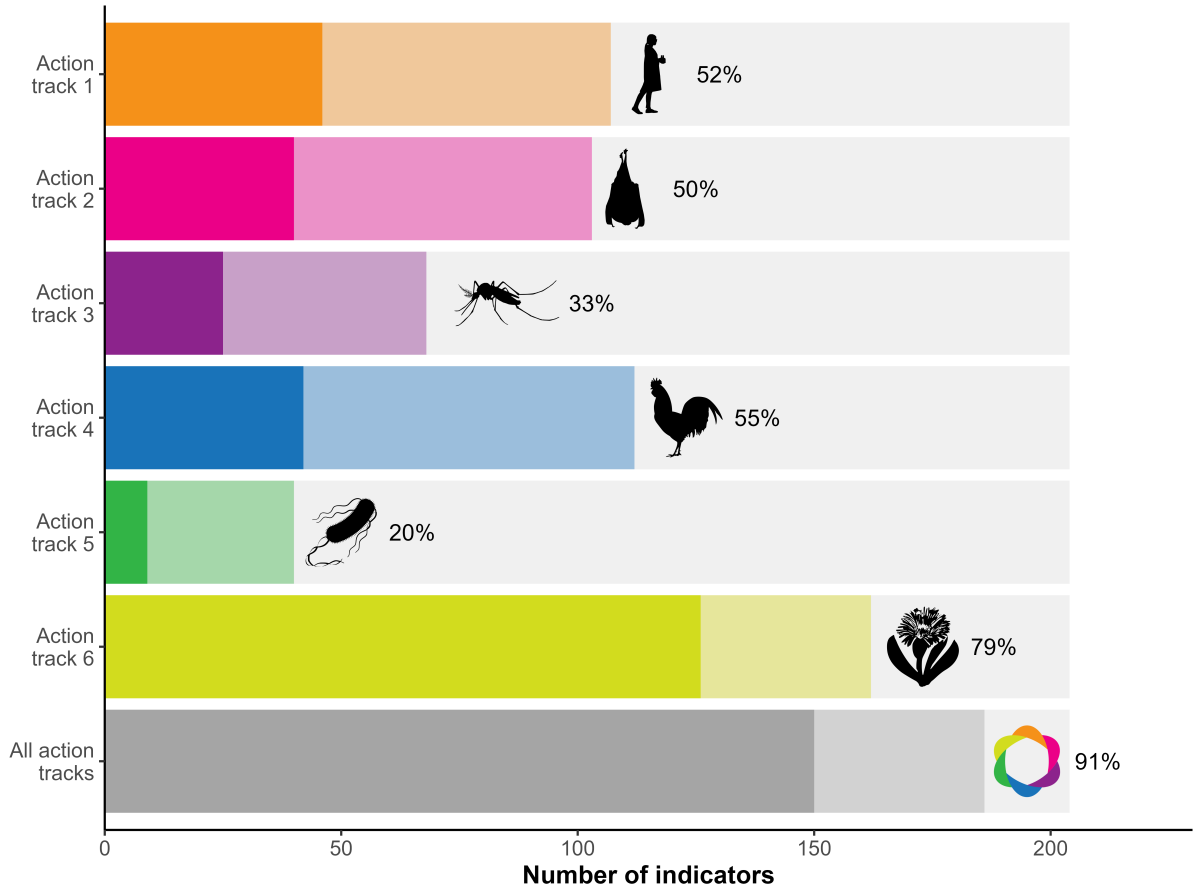

Figure 2: **Number of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks.** The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The proportion of usable indicators (either directly or after adaptation) for each action track is indicated. The bottom bar and proportion designate the indicators that can monitor at least one action in the whole plan.

- Gaps in indicators
  - Indicators
- Other indicators could be identified in other monitoring frameworks (e.g. SDG)
  - New indicators can be developed after identifying gaps

Conference of the Parties to the Convention on Biological Diversity. (2025). *Monitoring framework for the Kunming-Montreal Global Biodiversity Framework*. United Nations Environment Programme. Decision no. CBD/COP/16/L.26/Rev.1.

UNEP-WCMC. (2025). Indicators for the Kunming – Montreal Global Biodiversity Framework. In *Indicator Repository*. https://www.gbf-indicators.org/.

Table 1: **Number of KM-GBF indicators that can be used directly and after adaptation to monitor each action of the OH JPA.** The total number of usable indicators for each action track is indicated in the highlighted blue lines. A specific indicator can be used to monitor multiple action tracks, but only the most relevant action in each action track was identified.

Action	Directly usable	Usable after adaptation	Total usable
1 Enhancing One Health capacities to strengthen health systems	**46**	**61**	**107**
1.1 Establish the foundations for One Health capacities	30	5	35
1.2 Generate mechanisms, tools, and capacities to establish a One Health competent workforce and the frameworks/processes to facilitate One Health work	8	50	58
1.3 Generate an enabling environment for the effective implementation of One Health	8	6	14
2 Reducing the risks from emerging and re-emerging zoonotic epidemics and pandemics	**40**	**63**	**103**
2.1 Understand the drivers of emergence, spillover, and spread of zoonotic pathogens	16	27	43
2.2 Identify and prioritize targeted, evidence-based upstream interventions to prevent the emergence, spillover, and spread of zoonotic pathogens	7	6	13
2.3 Strengthen national, regional, and global One Health surveillance, early warning, and response systems	17	30	47
3 Controlling and eliminating endemic zoonotic, neglected tropical and vector-borne diseases	**25**	**43**	**68**
3.1 Enable countries to develop and implement community-centric and risk-based solutions to endemic zoonotic, neglected tropical, and vector-borne disease control using a One Health approach involving all relevant stakeholders	3	14	17
3.2 Ensure the harmonized application of One Health principles at all levels by implementing practical measures to strengthen local, national, regional, and global policy frameworks for the control and prevention of endemic zoonotic, neglected tropical, and vector-borne diseases	18	10	28
3.3 Increase political commitment and investment in the control of endemic zoonotic, neglected tropical, and vector-borne diseases, by advocating for and demonstrating the value of a One Health approach	4	19	23
4 Strengthening the assessment, management and communication of food safety risks	**42**	**70**	**112**
4.1 Strengthen the One Health approach in national food control systems and food safety coordination	18	30	48
4.2 Utilize and improve food systems data and analysis, scientific evidence, and risk assessment in developing policy and making integrated risk management decisions	15	21	36
4.3 Foster the adoption of the One Health approach in national foodborne disease surveillance systems and research for the detection and monitoring of foodborne disease and food contamination	9	19	28
5 Curbing the silent pandemic of antimicrobial resistance (AMR)	**9**	**31**	**40**
5.1 Strengthen the capacity and knowledge of countries to prioritize and implement context-specific collaborative One Health work to control AMR in policy, legislation, and practice	7	17	24
5.2 Reinforce global and regional initiatives and programmes to influence and support One Health responses to AMR	2	13	15
5.3 Strengthen global AMR governance structures	0	1	1
6 Integrating the environment into One Health	**126**	**36**	**162**
6.1 Protect, restore, and prevent the degradation of ecosystems and the wider environment	92	16	108
6.2 Mainstream the health of the environment and ecosystems into the One Health approach	9	7	16
6.3 Integrate environmental knowledge, data, and evidence into One Health decision-making	20	6	26
6.4 Create an interoperable One Health academic and in-service training programme for environmental, medical, agricultural, and veterinary sector professionals	5	7	12

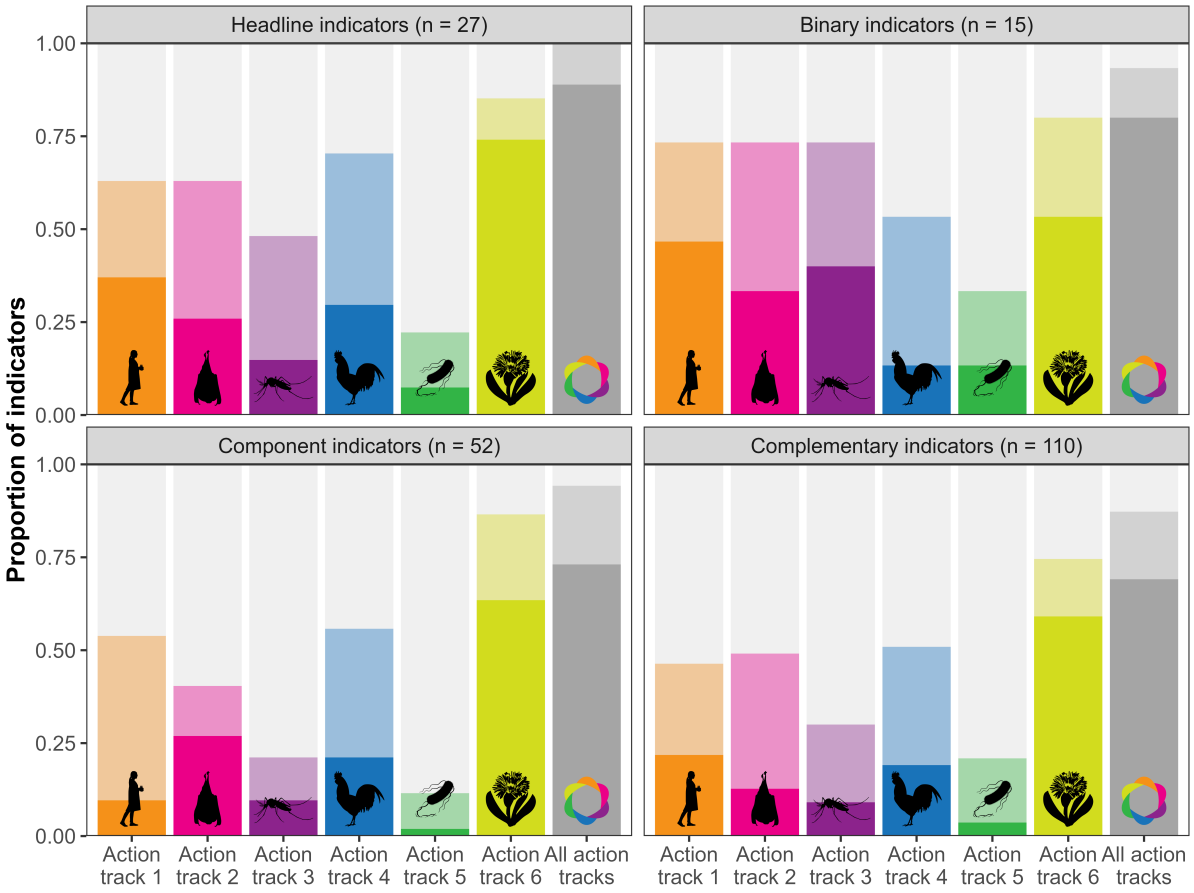

Figure 3: **Proportion of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks, partitioned by the different groups of indicators.** The number of indicators in each group is indicated in parentheses. The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The gray bars designate the indicators that can monitor at least one action in the whole plan.

Figure 4: **Proportion of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks, partitioned by the different thematic categories of the Global Action Plan on Biodiversity and Health.** The number of indicators whose target is in each thematic category is indicated in parentheses. The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The gray bars designate the indicators that can monitor at least one action in the whole plan. Indicators that monitor KM-GBF goals A, B, C, and D are not assigned to any thematic category.